

Capstone Modul 3

Hotel Booking Demand - Machine Learning Model

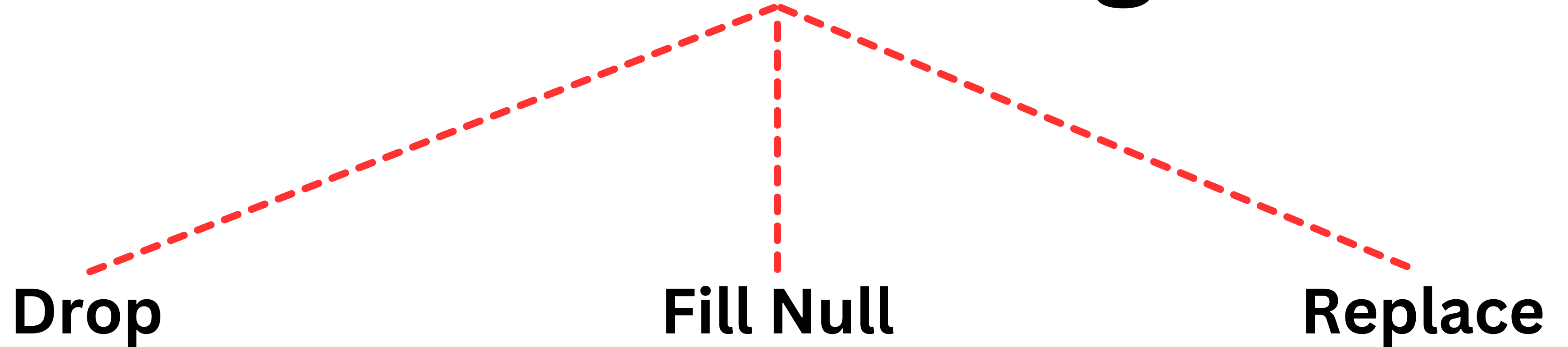
By: Benjamin Nikholas

Data Understanding

Business Problem

**ML Model that suits for predicting
customer booking cancelation**

Data Cleaning



Consistency and Outlier Check

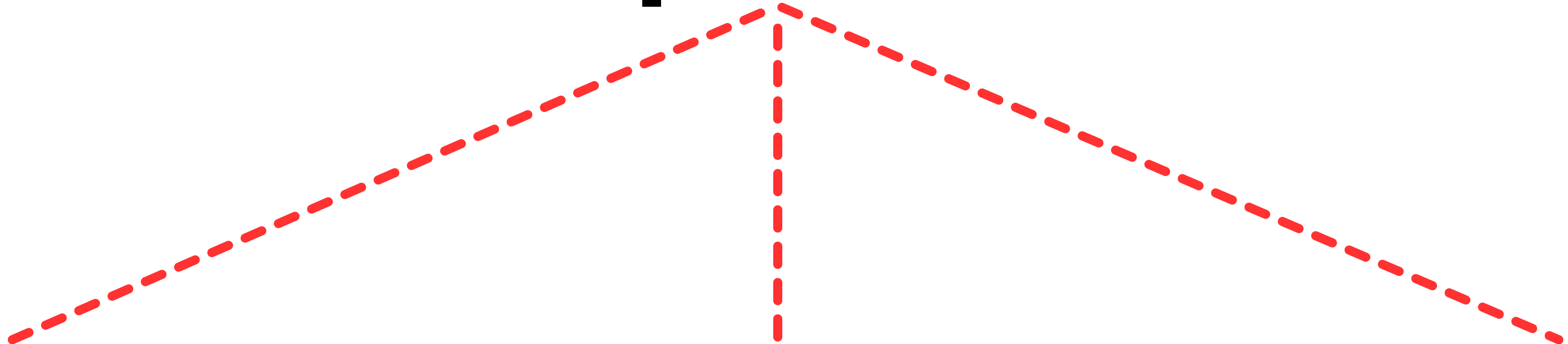
	Column	Lower Bound	Upper Bound	Unique Values	Outlier Count
0	adr	-15.775	211.065	8879	0
1	lead_time	-195.000	373.000	479	0
2	agent	-210.500	369.500	334	0
3	days_in_waiting_list	0.000	0.000	128	115692
4	previous_bookings_not_canceled	0.000	0.000	73	115770
5	arrival_date_week_number	-17.000	71.000	53	0
6	stays_in_week_nights	-2.000	6.000	35	0
7	arrival_date_day_of_month	-14.500	45.500	31	0
8	booking_changes	0.000	0.000	21	101314
9	stays_in_weekend_nights	-3.000	5.000	17	0
10	previous_cancellations	0.000	0.000	15	112906
11	adults	2.000	2.000	14	89680
12	total_of_special_requests	-1.500	2.500	6	0
13	babies	0.000	0.000	5	118473
14	children	0.000	0.000	5	110800
15	required_car_parking_spaces	0.000	0.000	5	111974
16	arrival_date_year	2014.500	2018.500	3	0
17	is_repeated_guest	0.000	0.000	2	115580

Pipeline

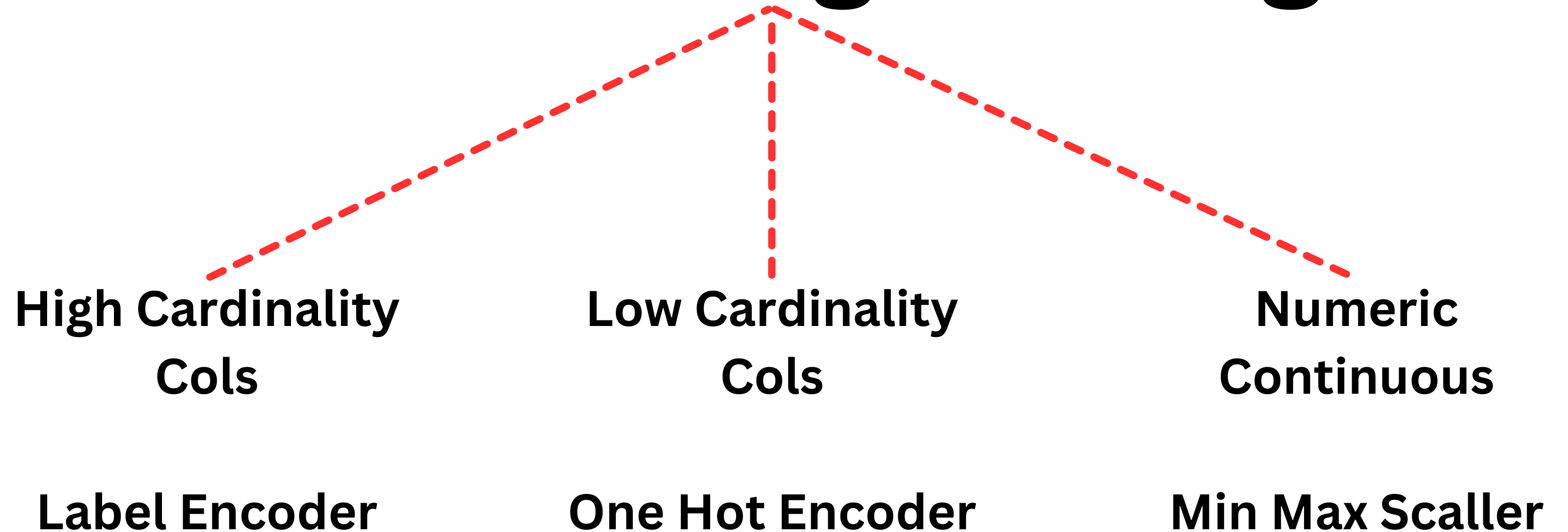
**Feature
Selection**

**Feature
Engineering**

**ML
Algorithms**



Feature Engineering



Feature Selection



SelectKBest

Score Function = ANOVA F-Value

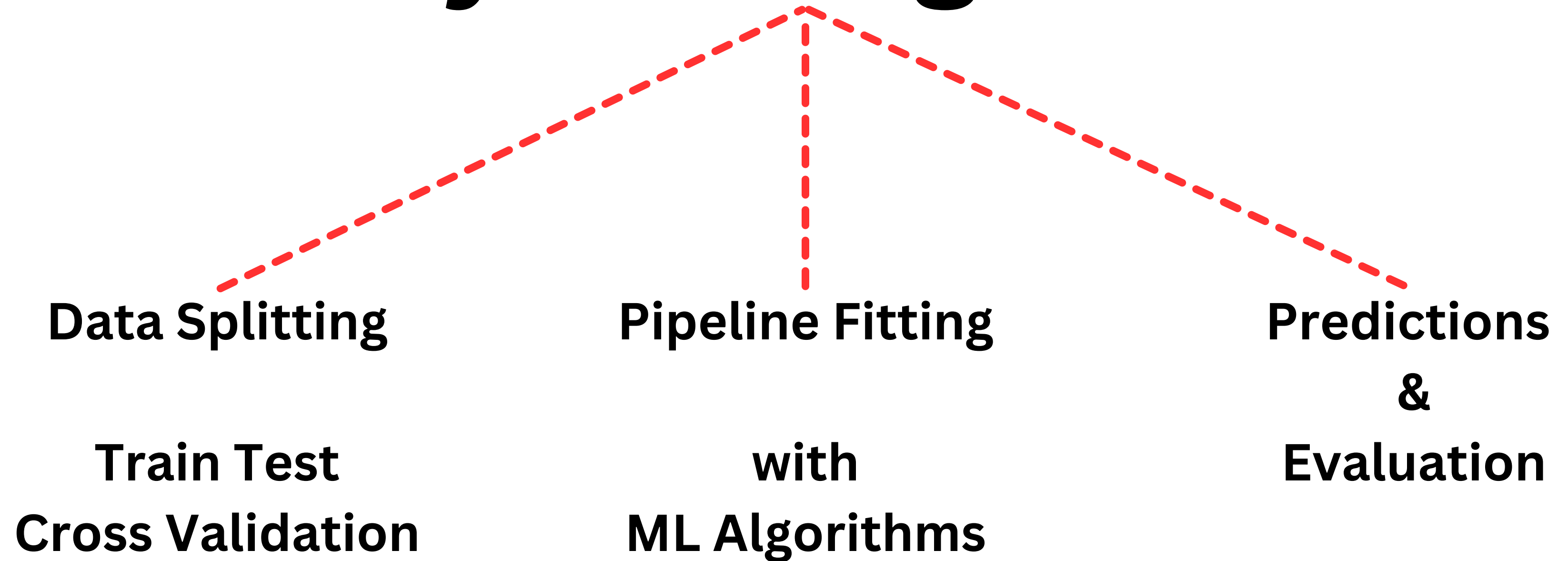
ML Algorithms



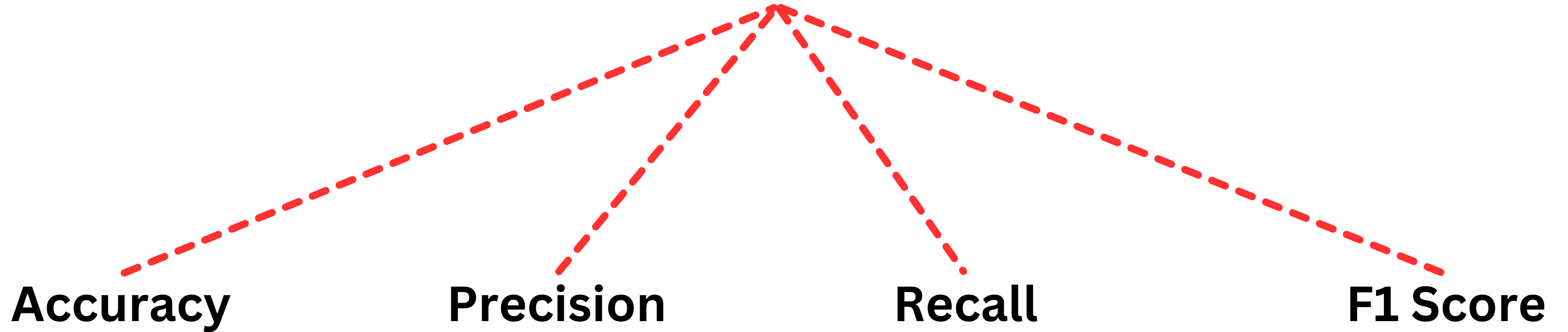
Binary Classification (Yes / No)

Logistic Regression

Analysis & Algorithms



Evaluation Metrics



Conclusion & Recommendation